

Module - 1 (Fundamental)

1. What is SDLC?

SDCL stands for **Software Development Life Cycle**. It is a **step-by- step process** used to develop software in a proper and organized way.

It helps in planning, designing, developing, testing, and maintaining software.

2. What is software testing?

Software testing is a process use to identify the corrections, completeness and quality of developed computer software.

3. What is agile methodology?

- A flexible way to build software in small parts.
- If focuses on teamwork and quick changes.

4. What is SRS?

- **SRS:** Software Requirement Specification.
- SRS is a written document that describes all the requirements of a software system.

5. What is OOPS?

OOPS (Object-Oriented Programming System) is a programming method based on objects and classes.

6. Basic Concepts of OOPS?

- Class
- Object
- Inheritance
- Encapsulation
- Polymorphism
- Abstraction

7. What is object?

- A real-world thing represented in code.
- Example: a car object

8. What is object?

A blueprint to create objects.

Example: car class to make car objects.

9. What is Encapsulation?

Encapsulation means **hiding data** and allowing access only through methods.

10. What is Inheritance?

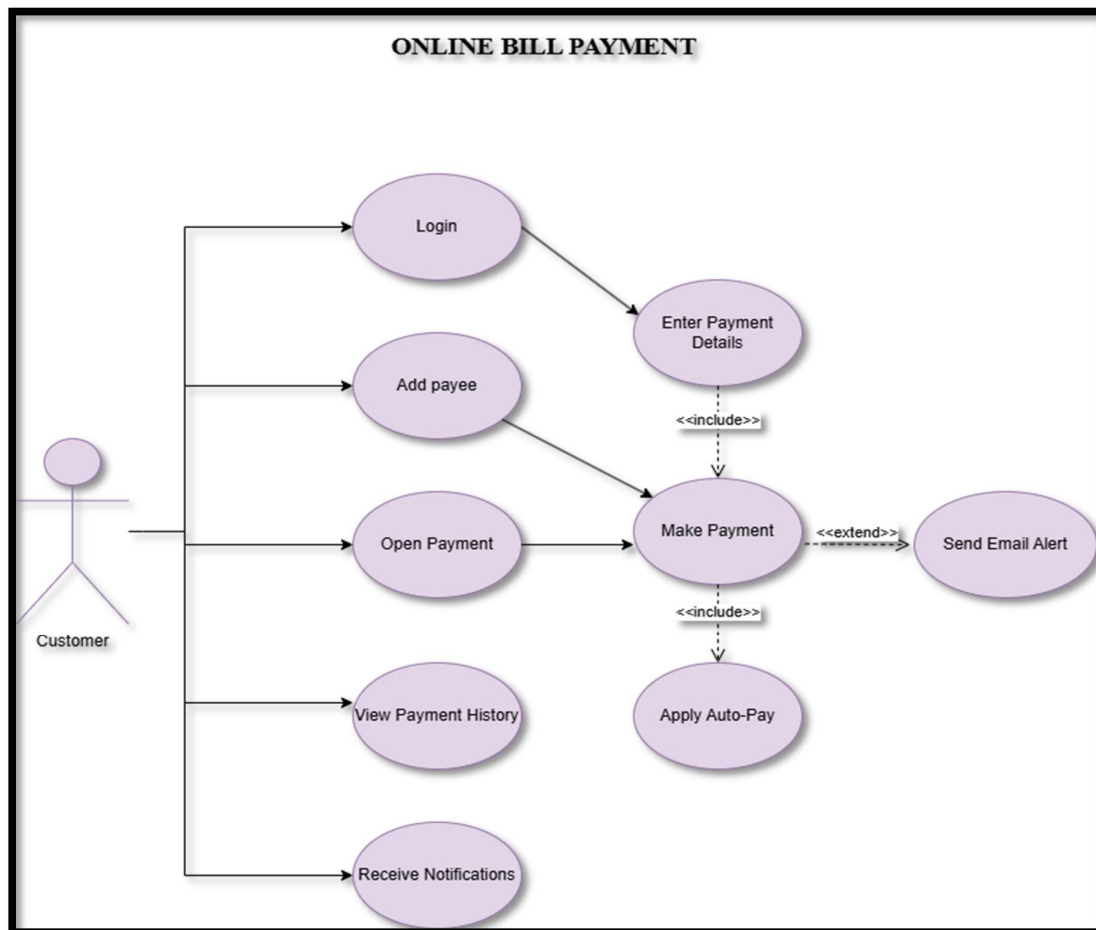
Inheritance means one class can use properties of another class.

Example: "Sportscar" inherits from "car".

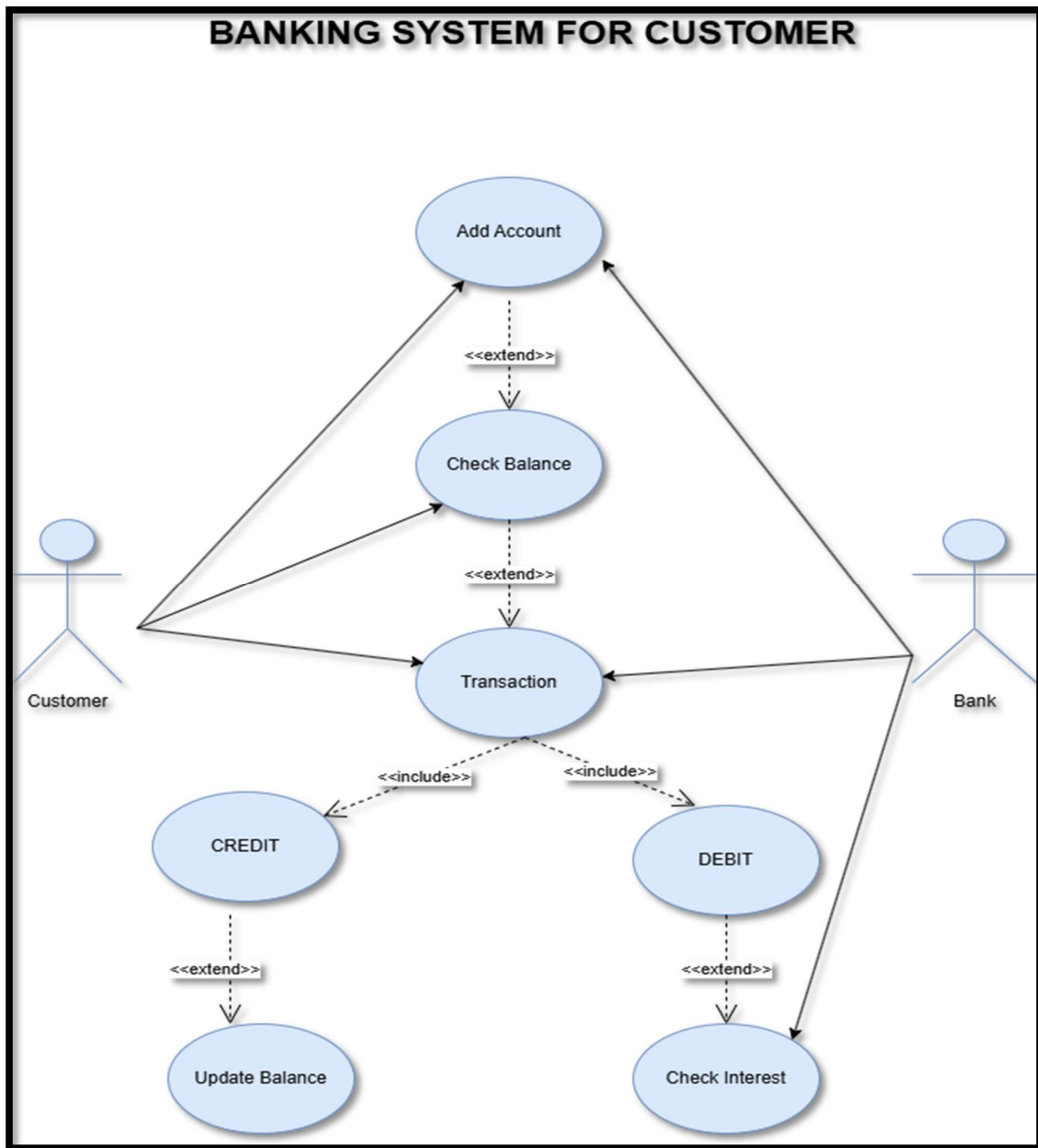
11. What is Polymorphism?

Polymorphism means one function, many forms (same method works differently).

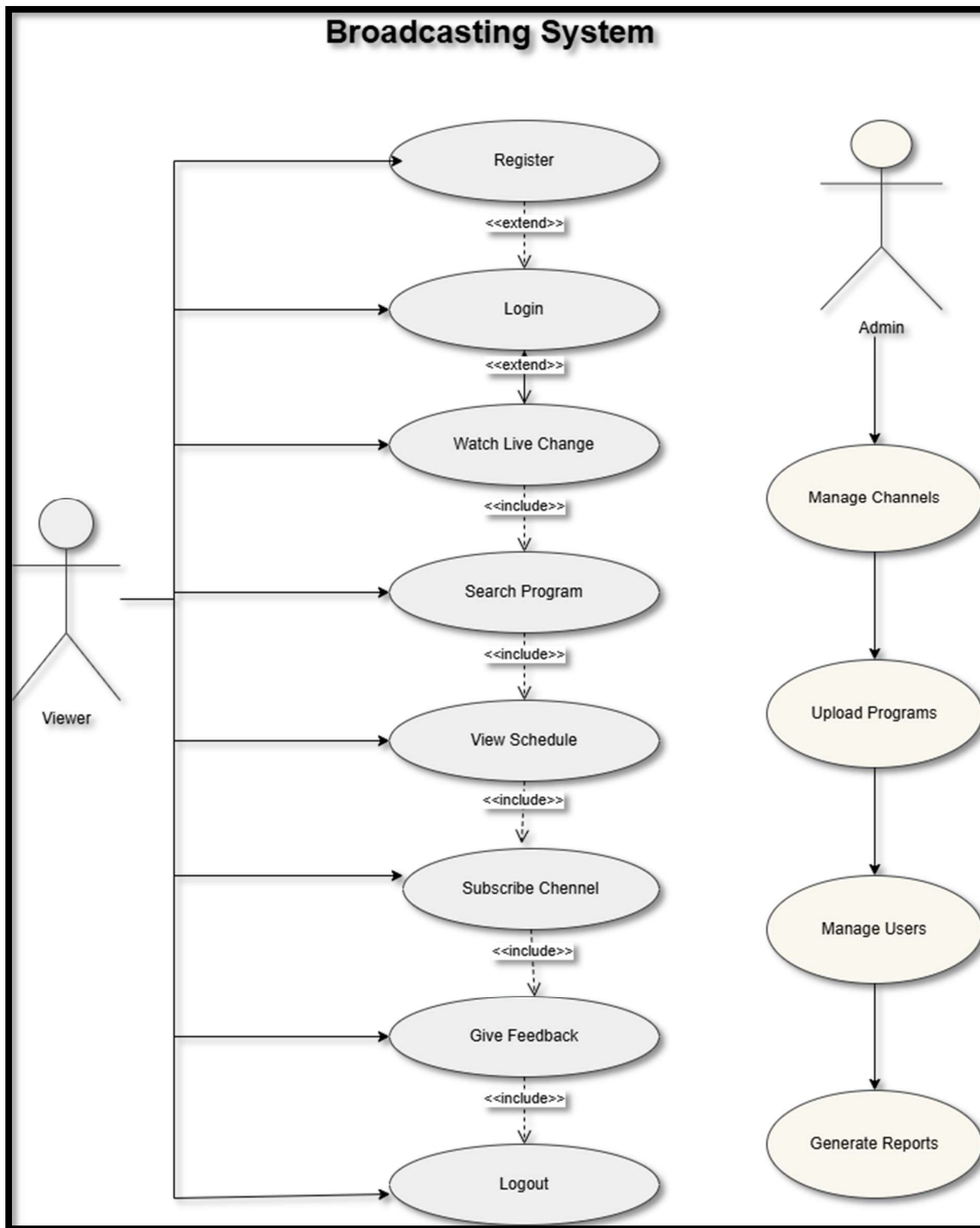
12. Draw Usecase on online bill payment system.



13. Draw Usecase on banking system for customers.



14. Draw Use case on Broadcasting System.



15. Write SDLC phases with basic introduction.

- **Requirement Gathering:** In this phase, developer collects all customer needs and software requirement.
- **Design:** In this phase, system design is prepared like database design, interface design, and architecture.
- **Implementation:** In this phase, actual coding of the software is done by programmers.
- **Testing:** In this phase, completed software is tested to find and fix errors or bug.
- **Deployment:** In this phase, completed software is installed and delivered to users.
- **Maintenance:** In this phase, software is updated, improved, and errors are fixed after delivery.

16. Explain Phases of the waterfall mod.

- **Requirement:** In this phase, all customer requirements are collected and understood.
- **Desing:** In this phase, software structure, database, and interface design are prepared.
- **Implementation:** In this phase, programmers write the code for the software.
- **Testing:** In this phase, software is tested to find and remove errors.
- **Deployment:** In this phase, completed software is delivered and installed for users.
- **Maintenance:** In this phase, software is maintained, updated, and bugs are fixed after delivery

17. Write phases of spiral model

- Planning
- Risk Analysis
- Engineering
- Evaluation

18. Write agile manifesto principles

- Customer satisfaction
- Frequent delivery
- Teamwork
- Flexibility
- Continuous improvement

19. Explain working methodology of agile model and also write pros and cons.

Working:

- Work in small parts
- Continuous feedback

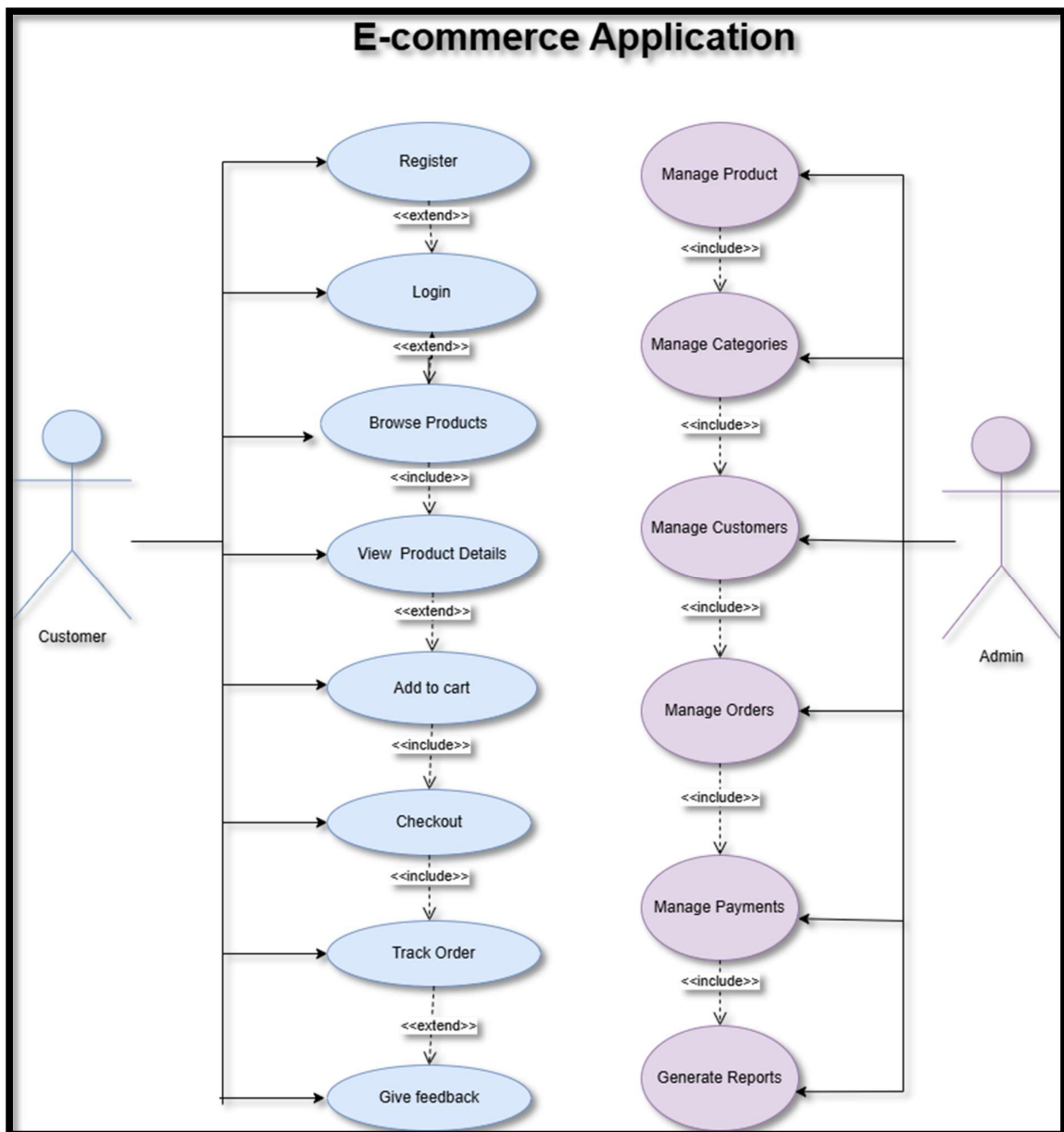
Advantages:

- Fast development
- Flexible
- Better quality

Disadvantages:

- Hard to manage large projects
- Needs Skilled team

20. Use Case – E-commerce Application



21. Use Case – Online Shopping with Payment Gateway

